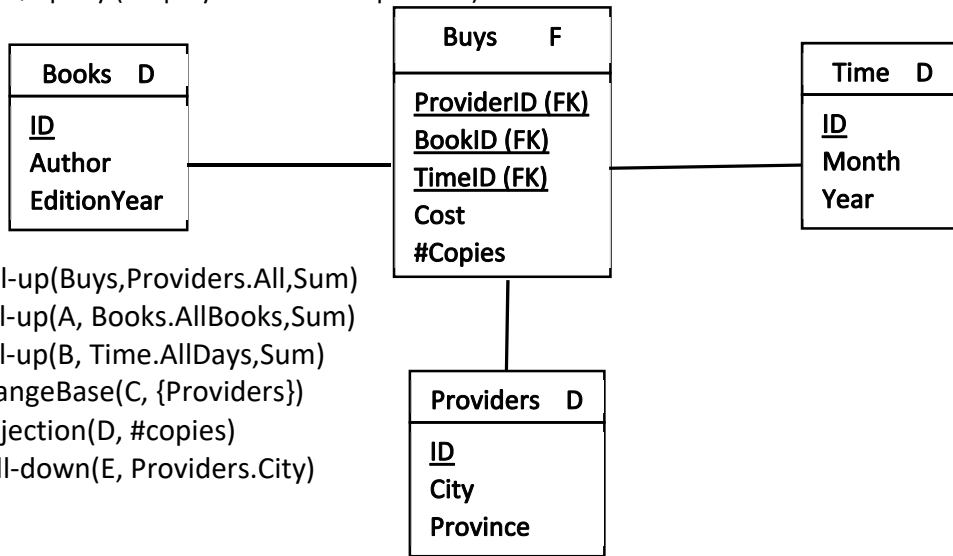


Surname: _____ Name: _____

1)(10%) Given the multidimensional schema and the sequence of **multidimensional operations** below, give the equivalent SQL query (simplify it as much as possible).



- A:=Roll-up(Buys,Providers.All,Sum)
- B:=Roll-up(A, Books.AllBooks,Sum)
- C:=Roll-up(B, Time.AllDays,Sum)
- D:=ChangeBase(C, {Providers})
- E:=Projection(D, #copies)
- R:=Drill-down(E, Providers.City)

2) (15%) Give the relational representation (i.e., tables and their corresponding integrity constraints) of the geographical **dimension** for a data mart in United Nations, so that it allows to **keep track of changes** in territories. Assume you only need to keep track of countries and their first administrative level (e.g., Autonomous communities in Spain). We would like to consider the split of countries (like in the case Catalonia would become independent), as well as annexations (like the case of Crimea peninsula by Russian Federation, assuming that peninsula was already a pre-existing component of Ukraine at the first administrative level). Briefly justify your answer and explicit any assumption you make.

3) (20%) Let's suppose we have a table T(Id, A, B, C) with 1,000 blocks and 100 records per block. We have a **materialized view** V(Id, A, B, C) with 100 blocks and also 100 records per block that contains a selection over T by a predicate involving A and B. Assuming that neither the table nor the materialized view have any order (i.e., they are not clustered), if we receive "UPDATE T SET C=C+1 WHERE Id%1000=0;" (being "%" the modulo operation) would it be better a complete or an incremental update of the view? Explicit any assumption you make and **numerically justify** your answer.

4) (10%) Consider the two relational schemas below (primary key attributes underlined). Would the instances of these schemas represent the **same reality**? Briefly explain why and exemplify it.

Flight(flightNum: String, origin: Airport, destination: Airport)
FlightEnd(flightNum: String, type: enum{Origin, Destination}, place: Airport)

5) (10%) Name three advantages of creating ETL processes **using ETL tools** (against hand-coded scripts).

- a. _____
- b. _____
- c. _____

6) (10%) Which ETL **process quality** characteristic is primarily improved by introducing Logical and Conceptual designs? Briefly explain how.

Surname: _____ Name: _____

7) (10%) Which logic property (or properties) of data quality rules is (are) violated by the given relational schema? Briefly explain why.

```
CREATE TABLE Flights (  
  flightID CHAR(26) NOT NULL,  
  departureAirport CHAR(3) NOT NULL REFERENCES Airports (ID),  
  arrivalAirport CHAR(3) NOT NULL REFERENCES Airports (ID)  
  actualDeparture TIMESTAMP WITHOUT TIME ZONE,  
  actualArrival TIMESTAMP WITHOUT TIME ZONE,  
  duration INTERVAL,  
  CONSTRAINT ckDepartureAirport CHECK (id <> 'BCN'),  
  CONSTRAINT ckArrivalAirport CHECK (id = 'BCN'),  
  CONSTRAINT ckMinDuration CHECK (duration > interval '02:30:00'),  
  CONSTRAINT ckMaxDuration CHECK (duration < interval '01:25:00')  
);
```

```
CREATE TABLE Airports (  
  id CHAR(3) PRIMARY KEY,  
  name VARCHAR(20) NOT NULL REFERENCES departments (ID),  
  CONSTRAINT ckAirport CHECK (id <> 'BCN')  
);
```

- 1) _____

- 2) _____

- 3) _____

- 4) _____
